1 Variational fermionic wave functions on lattices

1.1 Summary

1.2 Crash course on the origin of antisymmetric wave functions

This can be found in probably any introductory book on quantum mechanics, here I directly follow Schwabl, *Quantum Mechanics*, section 13.1.

The Heisenberg uncertainty relation implies that identical particles are indistinguishable. Consequently the probability amplitude $|\psi(x_1, \dots, x_N)|^2$ of a quantum many-body state is invariant under the exchange $x_i \leftrightarrow x_j$ of two indistinguishable particles. Define the exchange operator

$$P_{ij}\,\psi(x_1, \dots, x_i, \dots, x_j, \dots, x_N) \coloneqq \psi(x_1, \dots, x_j, \dots, x_i, \dots, x_N), \tag{1}$$

so that invariance of the amplitude gives

$$P_{ij}\,\psi(\boldsymbol{x}) = \lambda\,\psi(\boldsymbol{x}), \qquad |\lambda|^2 = 1. \tag{2}$$

Furthermore, exchanging the same two particles twice cannot change the wave function at all, hence

$$P_{ij}^2\,\psi(\boldsymbol{x}) = \lambda^2\psi(\boldsymbol{x}) = \psi(\boldsymbol{x}) \quad \Longrightarrow \quad \lambda^2 = 1, \text{ i.e. } \lambda = \pm 1. \tag{3}$$

Fermions are the particles for which $\lambda = -1$, i.e.

$$\boxed{\psi(\dots, x_i, \dots, x_j, \dots) = -\,\psi(\dots, x_j, \dots, x_i, \dots)} \tag{4}$$

and thus $\psi(\dots, x_i, \dots, x_i, \dots) = 0$.

1.3 Slater determinants

Here we loosely follow E. Koch, *Mean-Field Theory: Hartree-Fock and BCS* (Jülich 2024), section 1.

Let $\{\varphi_n\}$ be a complete, orthonormal single-particle basis,

$$\sum_n \varphi_n^*(x)\varphi_n(x') = \delta(x - x') \qquad \text{(complete)}, \tag{5}$$

$$\int dx\,\varphi_n^*(x)\varphi_m(x) = \delta_{nm} \qquad \text{(orthonormal)}. \tag{6}$$

An arbitrary N-particle function can then be expanded in its first argument as

$$a(x_1, \dots, x_N) = \sum_{n_1} a_{n_1}(x_2, \dots, x_N)\,\varphi_{n_1}(x_1), \qquad \text{with} \tag{7}$$

$$a_{n_1}(x_2, \dots, x_N) = \int dx_1'\,\varphi_{n_1}^*(x_1')\,a(x_1', x_2, \dots, x_N). \tag{8}$$

Iterating in all arguments, yields

$$\psi(x_1,\ldots,x_N) = \sum_{n_1,\ldots,n_N} a_{n_1,\ldots,n_N}\,\varphi_{n_1}(x_1)\cdots\varphi_{n_N}(x_N), \tag{9}$$

which is not antisymmetric yet. Now apply the antisymmetrizer $\mathcal{A}$, with $\mathrm{sgn}(P) = (-1)^P$:

$$\mathcal{A}\,\psi(x_1,\ldots,x_N) \overset{\text{[K (3)]}}{\equiv} \frac{1}{\sqrt{N!}}\sum_P (-1)^P\,\psi(x_{P(1)},\ldots,x_{P(N)}) \tag{10}$$

$$= \frac{1}{\sqrt{N!}}\sum_P (-1)^P \sum_{n_1,\ldots,n_N} a_{n_1,\ldots,n_N}\,\varphi_{n_1}(x_{P(1)})\cdots\varphi_{n_N}(x_{P(N)}) \tag{11}$$

$$= \sum_{n_1,\ldots,n_N} a_{n_1,\ldots,n_N}\,\Phi_{n_1,\ldots,n_N}(x_1,\ldots,x_N), \tag{12}$$

where we have used the *Slater determinant*

$$\Phi_{\alpha_1,\ldots,\alpha_N}(x_1,\ldots,x_N) \overset{\text{[K (4)]}}{\equiv} \frac{1}{\sqrt{N!}}\begin{vmatrix} \varphi_{\alpha_1}(x_1) & \varphi_{\alpha_2}(x_1) & \cdots & \varphi_{\alpha_N}(x_1) \\ \varphi_{\alpha_1}(x_2) & \varphi_{\alpha_2}(x_2) & \cdots & \varphi_{\alpha_N}(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_{\alpha_1}(x_N) & \varphi_{\alpha_2}(x_N) & \cdots & \varphi_{\alpha_N}(x_N) \end{vmatrix} \tag{13}$$

$$\overset{\text{Leibniz}}{=} \frac{1}{\sqrt{N!}}\sum_{P\in S_N} \mathrm{sgn}(P)\prod_{i=1}^{N}\varphi_{\alpha_i}\big(x_{P(i)}\big). \tag{14}$$

Hence **any** $\psi(x_1,\ldots,x_N)$ **can be expanded in Slater determinants.** Furthermore, it follows immediately that $\psi(\boldsymbol{x})$ vanishes if either $x_i = x_j$ or $\alpha_i = \alpha_j$ for $i \neq j$: **fermions cannot occupy the same position** x **or the same quantum state** α**.**

1.4 The variational principle

Following F. Becca & S. Sorella, *Quantum Monte Carlo Approaches for Correlated Systems*, section 1.4.

Formulate some variational state $|\psi_{\mathrm{var}}\rangle$ and consider

$$E_{\mathrm{var}} = \frac{\langle\psi_{\mathrm{var}}|H|\psi_{\mathrm{var}}\rangle}{\langle\psi_{\mathrm{var}}|\psi_{\mathrm{var}}\rangle} \qquad \text{[B\&S (1.38)]}. \tag{15}$$

Expanding in the energy eigenstates $|\varphi_i\rangle$,

$$|\psi_{\mathrm{var}}\rangle = \sum_i |\varphi_i\rangle\langle\varphi_i|\psi_{\mathrm{var}}\rangle = \sum_i a_i|\varphi_i\rangle \qquad \text{[B\&S (1.39)]}, \tag{16}$$

so that, using $\langle\varphi_i|\varphi_j\rangle = \delta_{ij}$,

$$E_{\mathrm{var}} = \Big(\sum_i |a_i|^2\Big)^{-1}\Big(\sum_i a_i|\varphi_i\rangle\Big)^\dagger H\Big(\sum_j a_j|\varphi_j\rangle\Big) = \Big(\sum_i |a_i|^2\Big)^{-1}\sum_i |a_i|^2 E_i. \tag{17}$$

Assume $E_i \leq E_j$ for $i < j$, i.e. E_0 is the ground-state energy. Then

$$E_{\text{var}} - E_0 = \Big(\underbrace{\sum_i |a_i|^2}_{>0} \Big)^{-1} \Big(\sum_{i \neq 0} \underbrace{|a_i|^2}_{\geq 0} \underbrace{(E_i - E_0)}_{\geq 0} \Big) \;\geq\; 0 \qquad \text{[B\&S (1.41)]}, \tag{18}$$

i.e. E_{var} **is an upper bound for the exact energy.** In practice: parametrize some $|\psi_{\text{var}}\rangle$ and minimize E_{var}.

1.5 Variational wave functions

1.5.1 Hartree-Fock wave functions

The Hartree-Fock (HF) approximation writes the many-body wave function as a product of single-particle orbitals,

$$\boxed{|\psi_{\text{HF}}\rangle = \prod_{\alpha=1}^{N_e} \phi_\alpha^\dagger |0\rangle, \qquad \phi_\alpha^\dagger = \sum_{i=1}^{N} \sum_\sigma W_{\sigma,\alpha,i}\, c_{i,\sigma}^\dagger, \quad \sigma = \uparrow, \downarrow} \tag{19}$$

with N lattice sites and N_e electrons. This leaves $2 \cdot N_e \cdot N$ parameters instead of the exponentially many amplitudes of a generic many-body state. The parameters are then optimized by minimizing

$$E_{\text{HF}} = \frac{\langle \psi_{\text{HF}} | H | \psi_{\text{HF}} \rangle}{\langle \psi_{\text{HF}} | \psi_{\text{HF}} \rangle}, \qquad \text{e.g.} \quad W_{\sigma,\alpha,i} \;\to\; W_{\sigma,\alpha,i} - \frac{\partial E_{\text{HF}}}{\partial W^*_{\sigma,\alpha,i}}\, \delta\tau \qquad \text{[B\&S (1.54)]}, \tag{20}$$

using Wirtinger derivatives (see **??**).

The HF wave function can equivalently be written in first quantization as **one Slater determinant**. Since it is a *product of single-particle orbitals*, it is mostly useful in the non-interacting case or at weak coupling. In the Hubbard model, turning up U makes the motion of one electron depend on where the other electrons are, i.e. it generates electron-electron correlation.

1.5.2 The Gutzwiller wave function

Gutzwiller's idea: describe the effect of the Hubbard U interaction by reducing the weight of configurations with multiply occupied sites. Start from some ground state $|\psi_0\rangle$ (e.g. non-interacting, a Slater determinant) and apply an operator P_g that suppresses such configurations:

$$|\psi_g\rangle = P_g |\psi_0\rangle \qquad \text{[B\&S (1.58)]} \tag{21}$$

$$\text{with} \qquad P_g = \exp\Big[-\frac{g}{2} \sum_{i=1}^{N} (n_i - n)^2 \Big] \qquad \text{[B\&S (1.59)]}, \tag{22}$$

where $n_i = \sum_\sigma n_{i\sigma}$ and $n = \frac{1}{N} \big\langle \sum_{i=1}^N n_i \big\rangle$ is the average density per site. Here g is the variational parameter ($g > 0$ for the repulsive Hubbard model).

Given some configuration $|x\rangle$, P_g counts how many sites within that state have higher or lower occupation than the average; for repulsive interactions these states are suppressed accordingly:

$$\langle x|\psi_g\rangle = P_g(x)\,\langle x|\psi_0\rangle, \qquad P_g(x) \leq 1 \qquad \text{[B\&S (1.60)]}. \tag{23}$$

Since $P_g(x) \geq 0$, it *does not* affect the sign structure.

In the Hubbard model a metal-insulator transition is expected at *finite* U/t for $n = 1$. The Gutzwiller wave function can only reproduce this for $g \to \infty$; otherwise there is always a non-zero probability for doublon-holon creation, and these could then move around freely and thus conduct.

Fully projected Gutzwiller wave function ($g \to \infty$).

Examine P_∞ for $n = 1$:

$$P_\infty = \lim_{g\to\infty} P_g = \lim_{g\to\infty} \exp\Big[-\frac{g}{2}\sum_{i=1}^{N}(\hat{n}_i - 1)^2\Big], \tag{24}$$

$$\begin{aligned} \Rightarrow \quad P_\infty|\psi\rangle &= \lim_{g\to\infty} \begin{cases} \exp\big(-\frac{g}{2}a\big), & \text{if } \exists \text{ site with a holon/doublon } (a>0) \\ \exp(0), & \text{else} \end{cases} |\psi\rangle \\ &= \begin{cases} 0 \\ |\psi\rangle \end{cases} \end{aligned} \tag{25}$$

Therefore we can rewrite P_∞ as (B&S (1.63))

$$P_\infty = \prod_i \big(n_{i,\uparrow} - n_{i,\downarrow}\big)^2. \tag{26}$$

If we want to allow empty sites, one has to set

$$P_\infty = \prod_i \big(1 - n_{i,\uparrow} n_{i,\downarrow}\big), \tag{27}$$

which is important for $n < 1$, since then holons and doublons do not come in pairs.

For $g = \infty$ no density fluctuations can occur, whereas in a true insulator density fluctuations do occur but produce no net current – e.g. double-occupancy fluctuations via virtual processes in the Hubbard model at large U[1].

1.5.3 The density-density Jastrow wave function

Since the Gutzwiller wave function does not correctly describe an insulator, the ansatz requires modification. We can, for instance, change the correlation term that is applied

[1] see *Lecture Notes on Correlation and Magnetism*, Patrik Fazekas, chapter 5, t-J model)

to $|\psi_0\rangle$ (or change $|\psi_0\rangle \rightsquigarrow |\psi_{U=\infty}\rangle$). The Jastrow factor does this by including long-range correlations:

$$\mathcal{J} = \exp\Big[-\frac{1}{2}\sum_{i,j} v_{ij}\,(n_i - n)(n_j - n)\Big] \qquad \text{[B\&S (1.65)]} \tag{28}$$

$$|\psi_{\mathcal{J}}\rangle = \mathcal{J}|\psi_0\rangle = |\psi_g\rangle, \quad \text{for } v_{ij} = \delta_{ij}\, g \qquad \text{[B\&S (1.64)]}. \tag{29}$$

For translationally invariant systems the variational parameters are $v_{ij} = v_{|i-j|}$, i.e. $\mathcal{O}(L)$ possible different distances.

The long-range correlations induced by $\mathcal{J}$ shall create a bound state between holons and doublons, possibly hindering conduction while still allowing local density fluctuations. It can be shown that the behaviour of v_{ij} resulting from an energy minimization is *not* enough to capture a fully gapped phase (power-law behaviour is present, but the Fourier transform of v_{ij} decays exponentially).

Jastrow-Slater wave function.

In the case $v_{ij} = v_{ji}$ the Jastrow factor is symmetric under particle exchange, so (in first quantization) it can be combined with a Slater determinant to obtain a suitable fermionic wave function, $\psi_{\mathrm{JS}}(\boldsymbol{x}) = \langle \boldsymbol{x}|\psi_{\mathrm{JS}}\rangle$:

$$\psi_{\mathrm{JS}}(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_{N_e}) = \exp\Big[-\sum_{i<j} v_{ij}(n_i - n)(n_j - n)\Big] \cdot \psi_{\mathrm{HF}}(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_{N_e}), \tag{30}$$

with $\psi_{\mathrm{HF}}(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_{N_e}) = \det\{\phi_\alpha(\boldsymbol{r}_i)\}$ and $\{\phi_\alpha(\boldsymbol{r}_i)\}$ a set of one-particle orbitals (see B&S (1.68)).

1.5.4 The backflow wave function

As explained earlier, the set of Slater determinants forms a basis of the N-electron Hilbert space, so one can systematically improve HF by adding more Slater determinants, but this is computationally very demanding. Instead: **try to improve one single Slater determinant by making the orbitals configuration dependent**, which goes beyond mean field.

Original idea.

The movement of one particle "pushes other particles away" in order to avoid large overlap among them, i.e. a "counter flow" of all other particles – hence *backflow correlations*. One therefore constructs the Slater determinant with modified coordinates (continuum backflow),

$$\boldsymbol{r}_i^{\,b} = \boldsymbol{r}_i + \sum_{j\neq i} \eta\big(|\boldsymbol{r}_i - \boldsymbol{r}_j|\big)\big(\boldsymbol{r}_i - \boldsymbol{r}_j\big), \tag{31}$$

which also describes the effective displacement of the i-th particle due to the j-th one.

On a lattice.

For lattice models this effective displacement is much less useful, since all distances are discretized. Instead one changes the one-particle orbitals:

$$\phi_k\big(\boldsymbol{r}^{\,b}_{i,\sigma}\big) \;\simeq\; \phi^b_k\big(\boldsymbol{r}_{i,\sigma}\big) \equiv \phi_k\big(\boldsymbol{r}_{i,\sigma}\big) + \sum_j c_{ij}[x]\,\phi_k\big(\boldsymbol{r}_{j,\sigma}\big) \tag{32}$$

where $\phi_k(\boldsymbol{r}_{i,\sigma}) = \langle 0|c_{i,\sigma}|\phi_k\rangle$, the $|\phi_k\rangle$ are eigenstates of the mean-field Hamiltonian, and $|x\rangle$ is the many-body configuration of the system.

This is essentially a *dynamic linear combination of orbitals*: dynamic, since it *depends on the electron configuration.* In the Hubbard model at large U one can hybridize the orbitals as follows:

$$\phi^b_k(\boldsymbol{r}_{i,\sigma}) = \varepsilon\,\phi_k(\boldsymbol{r}_{i,\sigma}) + \eta \sum_j t_{ij}\left(D_i H_j\right)\phi_k(\boldsymbol{r}_{j,\sigma}) \tag{33}$$

i.e. hybridize the orbitals at i and j if there is a doublon at i and a holon at j, such that the electron can minimize energy by going to the empty site[2].

Summary.

Backflow correlations improve a single Slater determinant by replacing fixed single-particle orbitals with **configuration-dependent orbitals**, effectively introducing **many-body corrections** through a determinant whose elements depend on the full configuration. Combining this with a Jastrow factor gives the Slater-Jastrow-backflow wave function.

1.6 Fermionic systems with determinants

In this subsection we review section 5.6 of the book *Quantum Monte Carlo Approaches for Correlated Systems* by Federico Becca and Sandro Sorella. The presentation closely follows Becca and Sorella, with additional details included for clarity.

Throughout this section, we assume a lattice that contains L sites and each site can contain one spin-up and one spin-down orbital. Therefore, we have $2L$ available orbitals that can be occupied by particles. We use one collective index that contains the site and the spin, by setting

$$\hat{c}_{i,\uparrow} \equiv \hat{d}_i \quad \text{and} \quad \hat{c}_{i,\downarrow} \equiv \hat{d}_{i+L}. \tag{34}$$

This corresponds to the fermionic-mode ordering $(1\uparrow, 2\uparrow, \ldots, L\uparrow, 1\downarrow, 2\downarrow, \ldots, L\downarrow)$. Any fixed ordering is fine, but it must be used consistently throughout.

[2] cf. *"Role of backflow correlations for the non-magnetic phase of the t-t′ Hubbard model"* by Tocchio and Sorella, Eqs. (2)-(4)

Any number-conserving quadratic fermionic Hamiltonian can be diagonalized:

$$\hat{H}_0 = \sum_{I,J=1}^{2L} t_{I,J} d_I^\dagger \hat{d}_J \tag{35}$$

$$= \boldsymbol{d}^\dagger \boldsymbol{T} \boldsymbol{d} = \boldsymbol{d}^\dagger \boldsymbol{U} \boldsymbol{U}^\dagger \boldsymbol{T} \boldsymbol{U} \boldsymbol{U}^\dagger \boldsymbol{d} \tag{36}$$

$$= \boldsymbol{\phi}^\dagger \boldsymbol{E} \boldsymbol{\phi} = \sum_{\alpha=1}^{2L} \varepsilon_\alpha \hat{\Phi}_\alpha^\dagger \hat{\Phi}_\alpha, \tag{37}$$

$$\text{with} \quad \phi_\alpha^\dagger = \sum_I U_{I,\alpha} d_I^\dagger. \tag{38}$$

The ground state is thus obtained by occupying the N_e lowest-energy single-particle orbitals:

$$|\Phi_0\rangle = \prod_{\alpha=1}^{N_e} \hat{\Phi}_\alpha^\dagger |0\rangle = \prod_{\alpha=1}^{N_e} \left(\sum_{I=1}^{2L} U_{I\alpha} \hat{d}_I^\dagger \right) |0\rangle . \tag{39}$$

The matrix U is in general a $2L \times 2L$ unitary matrix, where we only need the first N_e columns to construct the ground state, as can be seen above.

A general occupation-number configuration can be written as

$$|x\rangle = \hat{d}_{R_1}^\dagger \hat{d}_{R_2}^\dagger \cdots \hat{d}_{R_{N_e}}^\dagger |0\rangle , \tag{40}$$

where $R_1, \ldots, R_{N_e}$ are the occupied orbitals in the configuration x, with $R_j = 1, \ldots 2L$. This is of course a redundant labeling, since for example $c_1^\dagger c_2^\dagger |0\rangle = -c_2^\dagger c_1^\dagger |0\rangle$. But to describe arbitrary configurations, that can occur from applying the Hamiltonian to another arbitrary configuration, they are convenient. The resulting operator string need not be in canonical order and it then must be reordered, producing the appropriate fermionic sign. Thus, different orderings may represent the same physical configuration up to a sign. Therefore, we have

$$\begin{aligned} \langle x|y\rangle &= \langle 0| \hat{d}_{R_{N_e}} \cdots \hat{d}_{R_1} \hat{d}_{Q_1}^\dagger \cdots \hat{d}_{Q_{N_e}}^\dagger |0\rangle \\ &= \begin{cases} 0, & \text{iff } \{R_1, \ldots, R_{N_e}\} \neq \{Q_1, \ldots, Q_{N_e}\} \\ \text{sign}(p), & \text{else, with } p\left((Q_1, \ldots, Q_{N_e})\right) = (R_1, \ldots, R_{N_e}), \end{cases} \end{aligned} \tag{41}$$

where $\{.\}$ denotes a mathematical set and $(.)$ an ordered tuple.

The overlap between a general configuration $|x\rangle$ and the ground state of the quadratic Hamiltonian is thus given as

$$\begin{aligned} \langle x|\Phi_0\rangle &= \langle 0| \hat{d}_{R_{N_e}} \cdots \hat{d}_{R_1} \prod_{\alpha=1}^{N_e} \left(\sum_{i=1}^{2L} U_{i\alpha} \hat{d}_i^\dagger \right) |0\rangle \\ &= \langle 0| \hat{d}_{R_{N_e}} \cdots \hat{d}_{R_1} \left[\sum_{p \in P} \text{sign}(p) \prod_{\alpha=1}^{N_e} U_{p\{Q_j\},\alpha} \right] \hat{d}_{Q_1}^\dagger \cdots \hat{d}_{Q_{N_e}}^\dagger |0\rangle , \end{aligned} \tag{42}$$

with P containing all possible permutations of the Q_j. These have now to be restricted on all valid permutations of the R_j within x (see eq. (41)), which yields

$$\langle x|\Phi_0\rangle = \langle 0|\, \hat{d}_{R_{N_e}} \cdots \hat{d}_{R_1} \left[\sum_{p\in P} \mathrm{sign}(p) \prod_{\alpha=1}^{N_e} U_{R_{p\{Q_j\}},\alpha}\right] \hat{d}^\dagger_{R_1} \cdots \hat{d}^\dagger_{R_{N_e}} |0\rangle \tag{43}$$

$$= \sum_{p\in P} \mathrm{sign}(p) \prod_{\alpha=1}^{N_e} U_{R_{p\{Q_j\}},\alpha} \tag{44}$$

$$= \det\left[U_{R_i,\alpha}\right]_{\substack{i=1,\ldots,N_e \\ \alpha=1,\ldots,N_e}}, \tag{45}$$

where we used the Leibniz formula in the last step.

In other words, although U is a $2L \times 2L$ matrix, only the N_e columns corresponding to the occupied lowest-energy orbitals and the N_e rows corresponding to the occupied orbitals R_i of the configuration x are needed:

$$U_x = \begin{pmatrix} U_{R_1,1} & \cdots & U_{R_1,N_e} \\ \vdots & \ddots & \vdots \\ U_{R_{N_e},1} & \cdots & U_{R_{N_e},N_e} \end{pmatrix} \tag{46}$$

and hence

$$\langle x|\Phi_0\rangle = \det(U_x). \tag{47}$$

This construction can also be used for BCS-type Hamiltonians (see end of section 5.6.1 of the book by Becca and Sorella).

Variational Monte Carlo

Let's say we have some general fermionic Hamiltonian $\hat{H}$, which might not be quadratic and we choose the Slater determinant as our variational ansatz $\langle x|\Phi_U\rangle = \det(U_x)$ with the entries of the full U as the variational parameters. The variational Monte Carlo (VMC) workflow then roughly looks like

1. Sample fermionic occupation configurations $x = (R_1, \ldots, R_{N_e})$ following the distribution resulting from $\Phi_U(x) = \det(U_x)$.
2. Evaluate Hamiltonian matrix elements $\langle x|\, \hat{H}\, |x'\rangle$ and estimate the local energy $e_L(x) = \sum_{x'} \langle x|\, \hat{H}\, |x'\rangle \frac{\langle x'|\Phi_U\rangle}{\langle x|\Phi_U\rangle}$.
3. Compute the gradient of the energy $\langle E\rangle \approx \frac{1}{N}\sum_{n=1}^{N} e_L(x_n)$ and optimize the variational parameters with it.
4. Repeat until convergence.

The Jordan-Wigner mapping

The Jordan-Wigner mapping (JWM) does not change anything in the derivation above since it is only a *rewriting of the operators* (and besides doing the above derivation with the JWM

would probably be much more tedious). But it might come in handy when computing matrix elements such as $\langle x|\,\hat{H}\,|x'\rangle$. To see what changes in the VMC workflow described above, we only have to look at how the form of the ansatz $\Phi_U(x)$ might change, since we know that the fermionic and spin configurations map onto each other (see **??**). But again, since the JWM is simply a rewriting of operators, the overlap of two states stays exactly the same and therefore the determinant ansatz remains. **Hence, the only significant change in the VMC workflow is that the Hamiltonian has to be rewritten but the ansatz remains unchanged.**

Computing matrix elements as for instance $\langle x|\,\hat{H}\,|x'\rangle$ involves commuting fermionic operators, which necessitates some book keeping for the resulting sign. This book keeping is exactly what is done by the Jordan-Wigner strings. Thus, in practice or numerically, we often use the Jordan-Wigner in order to circumvent having to do this book keeping by hand.

1.7 Parametrization via neural networks

Now let us have a look at how neural networks are used for variational states. There is no single resource that reviews the physics-motivated approaches like backflow, hidden fermions etc., so we mostly look at the corresponding papers.

1.7.1 Backflow transformations via neural networks

Following D. Luo & B.K. Clark, *Backflow Transformations via Neural Networks for Quantum Many-Body Wave Functions*.

Recall the definition of backflow introduced by Tocchio et al. in "Role of backflow correlations for the non-magnetic phase of the t-t' Hubbard model", Eq. (4):

$$\phi^b_{k\sigma}\big(r_{i,\sigma};\boldsymbol{x}\big) = \phi_{k\sigma}\big(r_{i,\sigma}\big) + \sum_j \eta_{ij,\sigma}\,\phi_{k\sigma}\big(r_{j,\sigma}\big), \tag{48}$$

$$\eta_{ij,\sigma} = t\,D_i H_j\,\Theta_{|i-j|,\sigma}, \qquad \text{(L\&C (6))} \tag{49}$$

with $D_i = n_{i,\uparrow}n_{i,\downarrow}$ and $H_i = (1-n_{i,\uparrow})(1-n_{i,\downarrow})$ counting doublons and holons.

The i,j-dependence of the hopping amplitudes t_{ij} is absorbed in the variational parameters $\Theta_{|i-j|,\sigma}$; e.g. for next-nearest-neighbour hopping one sets them to zero if $|i-j| \neq 1, 2$.

Now we make the ansatz to parametrize the backflow part via a neural network:

$$\phi^b_{k\sigma}\big(r_{i,\sigma};\boldsymbol{x}\big) = \phi_{k\sigma}\big(r_{i,\sigma}\big) + a^{\mathrm{NN}}_{ki,\sigma}(\boldsymbol{x}) \qquad \text{(L\&C (5))} \tag{50}$$

The typical advantage of a NN parametrization is that neural networks are universal approximators. On top of that, one can show **explicitly how to construct the backflow with a NN with two hidden layers**:

Step 1: rewrite η as a ReLU.

Assuming $t\,\Theta_{|i-j|,\sigma} \geq 0$,

$$\eta_{ij,\sigma} = t\,D_i H_j\,\Theta_{|i-j|,\sigma} = t\,n_{i,\uparrow}n_{i,\downarrow}h_{j,\uparrow}h_{j,\downarrow}\,\Theta_{|i-j|,\sigma},$$

which is nonzero only if $n_{i,\uparrow} = n_{i,\downarrow} = h_{j,\uparrow} = h_{j,\downarrow} = 1$, and hence

$$\eta_{ij,\sigma} = \mathrm{ReLU}\Big[t\,\Theta_{|i-j|,\sigma}\big(n_{i,\uparrow} + n_{i,\downarrow} + h_{j,\uparrow} + h_{j,\downarrow} - 3\big)\Big].$$

Step 2: input layer.

The NN shall have the input layer

$$\big(\underbrace{\sigma_1, \ldots, \sigma_N}_{\text{spin up}}, \underbrace{\sigma_{N+1}, \ldots, \sigma_{2N}}_{\text{spin down}}\big), \qquad \text{with}$$

$$\sigma_i = 2n_{i,\uparrow} - 1, \quad \sigma_{i+N} = 2n_{i,\downarrow} - 1, \quad i = 1, \ldots, N.$$

Inverting the relation for σ_i and inserting it into $\eta_{ij,\sigma}$,

$$\eta_{ij,\sigma} = \mathrm{ReLU}\Big[\tfrac{1}{2}\,t\,\Theta_{|i-j|,\sigma}\big(\sigma_i + \sigma_{i+N} - \sigma_j - \sigma_{j+N} - 2\big)\Big],$$

with $i, j \in \{1, \ldots, N\}$ and $\sigma = \pm 1$.

Step 3: first hidden layer.

Use this to construct $\eta_{ij,\sigma}$ as the following NN (cf. fig. 1):

$$v_k = \sum_{\ell=1}^{2N} w_{k\ell}\,\sigma_\ell + b_k, \qquad \text{set} \quad k = \sigma \cdot (i + N \cdot j). \tag{51}$$

$$\Rightarrow \quad v_{\sigma\cdot(i+N\cdot j)} = \sum_{\ell=1}^{2N} w_{\sigma\cdot(i+N\cdot j),\ell}\,\sigma_\ell + b_{\sigma\cdot(i+N\cdot j)}, \tag{52}$$

with

$$b_{\sigma\cdot(i+N\cdot j)} = -t\,\Theta_{|i-j|,\sigma} \qquad \text{and} \qquad w_{\sigma\cdot(i+N\cdot j),\ell} = \frac{t}{2}\,\Theta_{|i-j|,\sigma} \begin{cases} 1 & \text{if } \ell \,\%\, N = i \\ -1 & \text{if } \ell \,\%\, N = j \\ 0 & \text{else} \end{cases} \tag{53}$$

such that $\eta_{ij,\sigma} = \mathrm{ReLU}\big(v_{\sigma\cdot(i+N\cdot j)}\big)$.

Step 4: second hidden layer.

Finally, the full backflow is obtained by adding another hidden layer which takes the $\eta_{ij,\sigma}$ as inputs and weights them by the single-particle orbitals $\phi_k(r_{j,\sigma})$. The activation function of this layer is simply the identity. If $t\,\Theta_{|i-j|,\sigma} < 0$, the minus sign can be absorbed into this layer, resulting in

$$a^{\mathrm{NN}}_{ki,\sigma}(\boldsymbol{r}) = \sum_j \underbrace{\eta_{ij,\sigma}}_{\text{inputs}}\,\underbrace{s_{ij,\sigma}\,\phi_{k\sigma}(r_{j,\sigma})}_{\text{weights}}, \qquad s_{ij,\sigma} = \mathrm{sgn}\big(t\,\Theta_{|i-j|,\sigma}\big) = \pm 1. \tag{54}$$

$\Rightarrow$ the backflow can be represented by a neural network.

In practice, one simply chooses some NN (RBM, MLP, CNN, ...) and lets backpropagation take care of the rest.

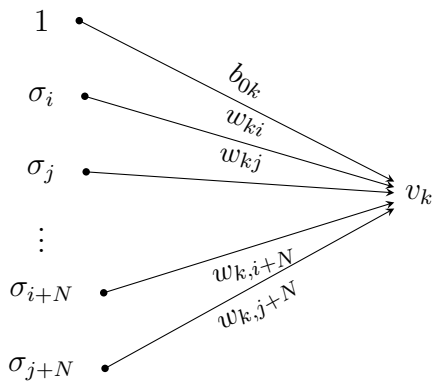


Figure 1: First hidden layer producing v_k, with $k = \sigma \cdot (i + N \cdot j)$.

1.8 References

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